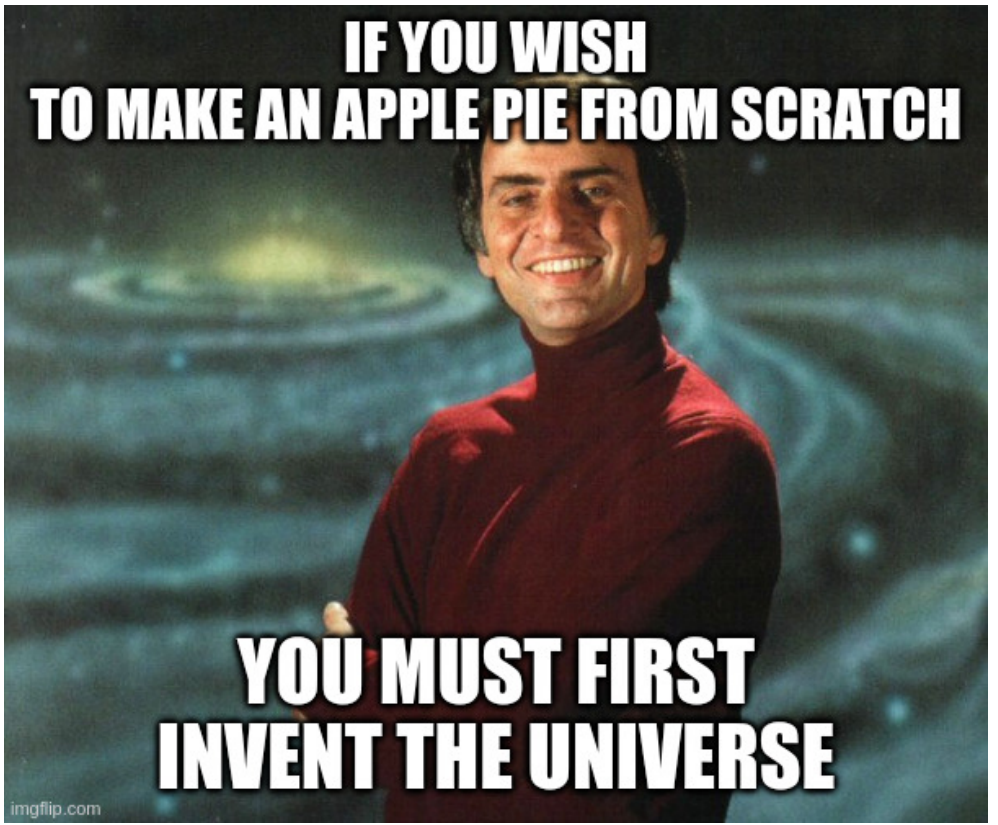
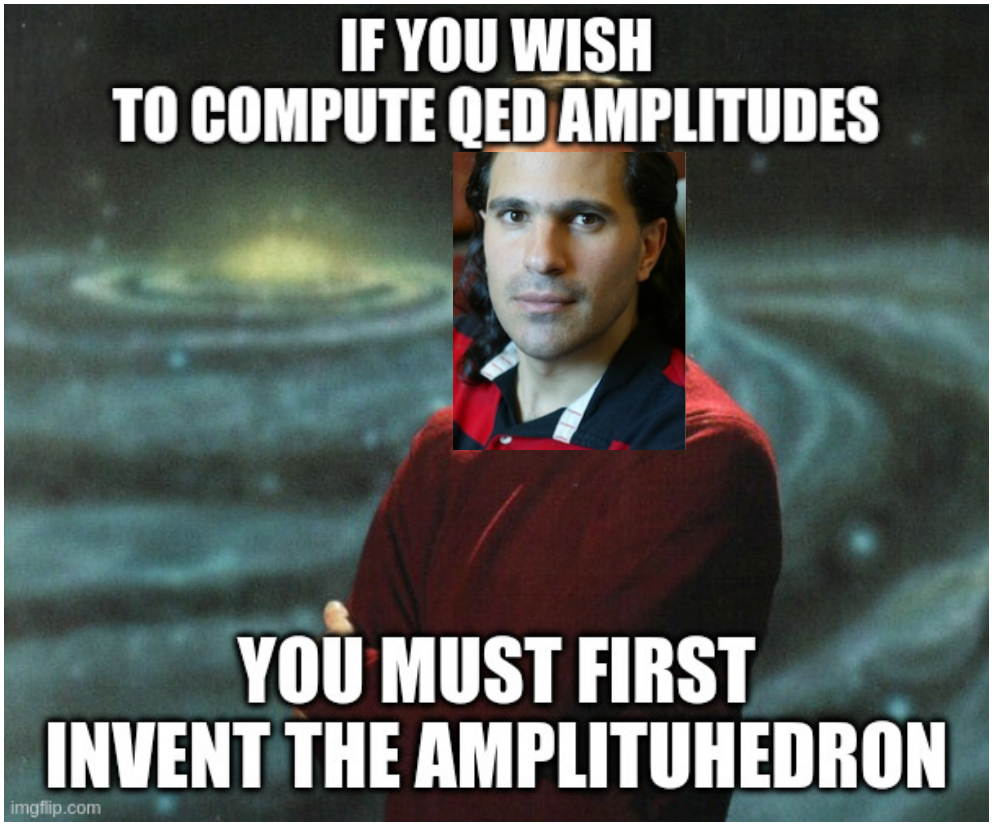
Michael Plesser
plesser@psu.edu

QED from $\mathcal{N} = 4$





$\mathcal{Q}$ Can $\mathcal{N} = 4$ help us compute (massless) QED amplitudes?

n.b. we've gotten *very* good at computing $\mathcal{N} = 4$ tree amps

$\mathcal{A}$ Surprisingly, yes!

Why is this surprising?

- Abelian vs non-Abelian theories,
- Gluons self-interact while photons do not,
- Different particle content in their spectrums,
- Lack of colour-ordering / partial amplitudes in QED

$\mathcal{Q}$ Can $\mathcal{N} = 4$ help us compute (massless) QCD amplitudes?

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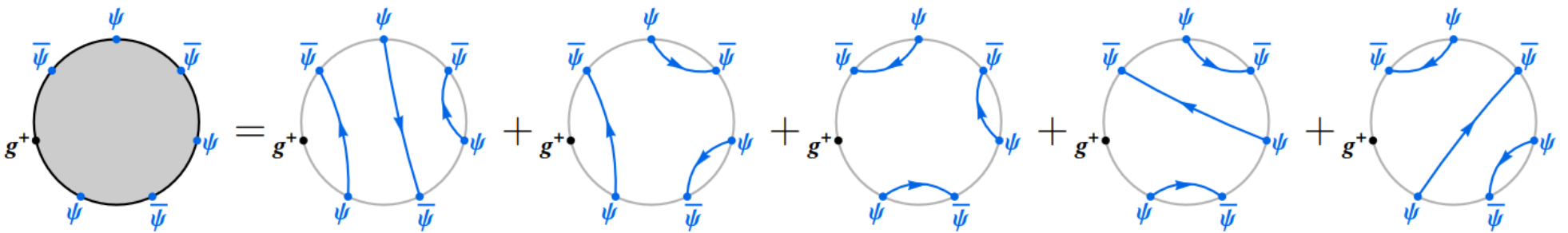
Why is this surprising?

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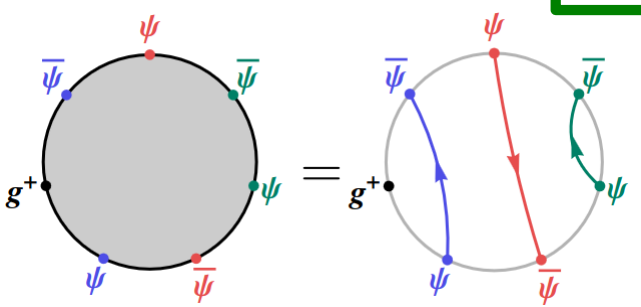
$\mathcal{N} = 4$ has scalars, QCD doesn't,
the number of fermion flavours can differ

One flavour is all you need (**Melia 1312.0599**)

QCD partial amps → sums of non-crossing chord diagrams



n.b. 1-flavour amplitudes are *identical* in QCD and $\mathcal{N} = 4$

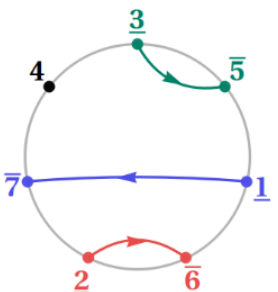


Compatibility of flavour-pairings and cyclic-ordering restrict what chord diagrams can contribute to a QCD partial amp

One flavour is all you need (**Melia 1312.0599**)

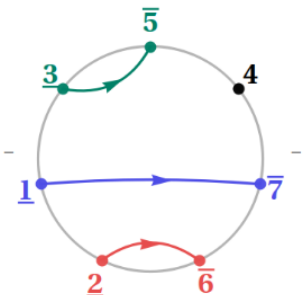
KK-relations let us *anchor* two legs and *orient* all lines.
→ "All-plus" basis of KK-independent partial amplitudes

$\mathcal{A}_3^1(\underline{2},\overline{7},4,\underline{3},\overline{5},\underline{1},\overline{6})$ =

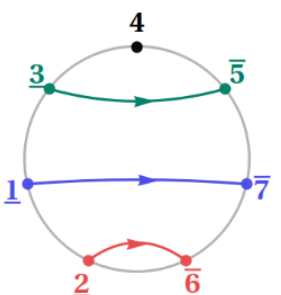


$$A(1,A__,\beta,B__,\alpha,C__) = (-1)^{1+|B|}\sum_{A=(a_L,a_R)}\ \sum_{\vec{\sigma}\in[(a_R\sqcup\overline{B})\beta]\sqcup C} A(1\,a_L\,\alpha\,\vec{\sigma})$$

=



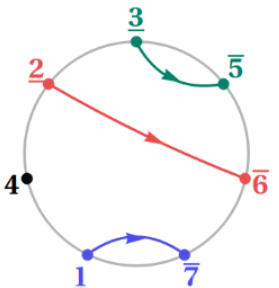
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One flavour is all you need (**Melia 1312.0599**)

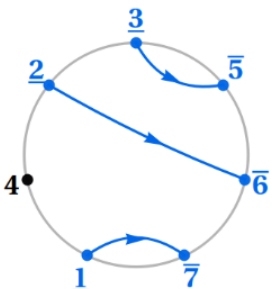
Some all-plus amplitudes are the same irrespective of N_f, those with chord diagrams of "highest-maturity"

$\mathcal{A}_3^1(\underline{1},4,\underline{2},\underline{3},\overline{\underline{5}},\overline{\underline{6}},\overline{\underline{7}})$

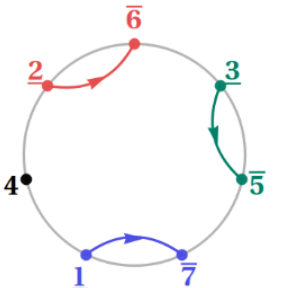


$=$

$A_3^1(\underline{1},4,\underline{2},\underline{3},\overline{\underline{5}},\overline{\underline{6}},\overline{\underline{7}})$

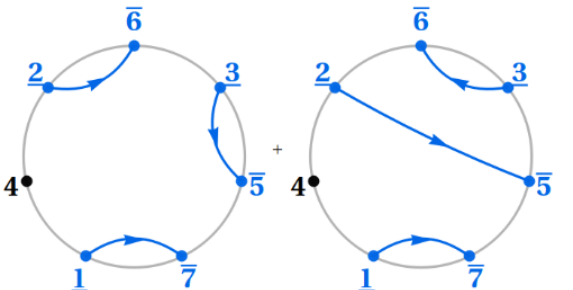


$\mathcal{A}_3^1(\underline{1},4,\underline{2},\overline{\underline{6}},\underline{3},\overline{\underline{5}},\overline{\underline{7}})$



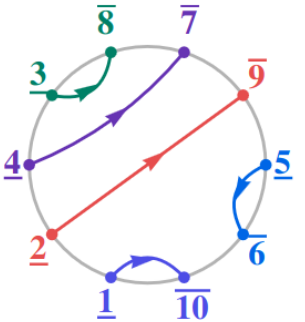
$\neq$

$A_3^1(\underline{1},4,\underline{2},\overline{\underline{6}},\underline{3},\overline{\underline{5}},\overline{\underline{7}})$

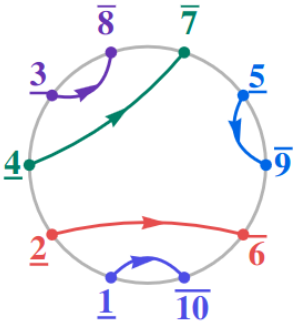


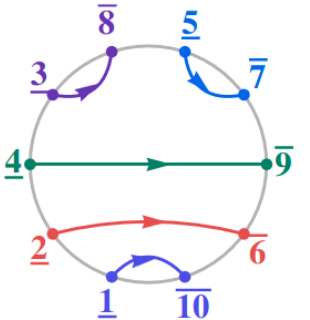
One flavour is all you need (**Melia 1312.0599**)

Melia's algorithm



$$\mathcal{A}(1243\bar{8}\bar{7}95\bar{6}\bar{1}\bar{0})$$





$$A(1243\bar{8}\bar{7}95\bar{6}\bar{1}\bar{0})+A(1254\bar{3}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})+\mathcal{A}(1243\bar{8}75\bar{9}\bar{6}\bar{1}\bar{0})$$

Highest maturity

$$A(1243\bar{8}75\bar{9}\bar{6}\bar{1}\bar{0})+A(1245\bar{3}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})+\mathcal{A}(1243\bar{8}5\bar{7}9\bar{6}\bar{1}\bar{0})$$

Highest maturity

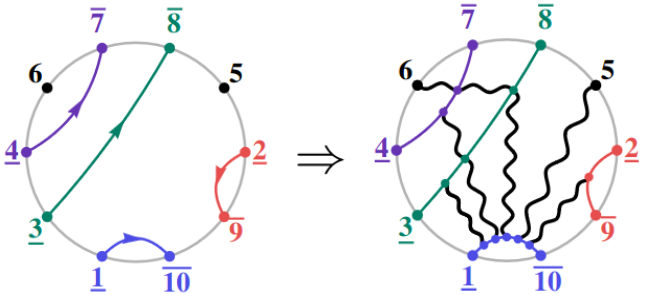
$$A(1243\bar{8}5\bar{7}9\bar{6}\bar{1}\bar{0})+A(1243\bar{5}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})$$

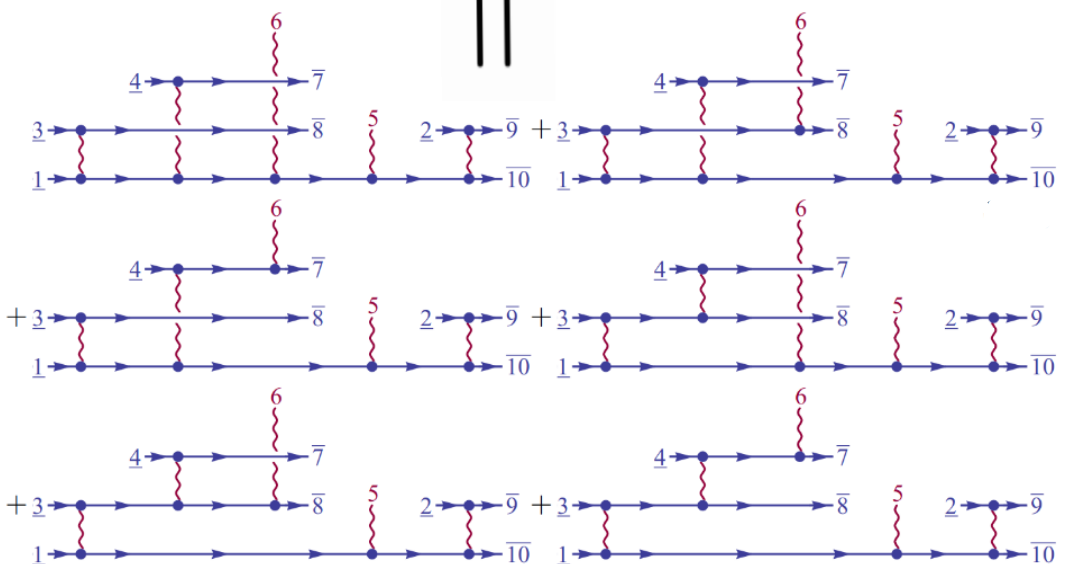
Highest maturity

$$\mathcal{A}(1243\bar{8}\bar{7}95\bar{6}\bar{1}\bar{0}) = A(1243\bar{8}\bar{7}95\bar{6}\bar{1}\bar{0})+A(1254\bar{3}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})+A(1243\bar{8}75\bar{9}\bar{6}\bar{1}\bar{0})$$
$$+A(1245\bar{3}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})+A(1243\bar{8}5\bar{7}9\bar{6}\bar{1}\bar{0})+A(1243\bar{5}\bar{8}\bar{7}9\bar{6}\bar{1}\bar{0})$$

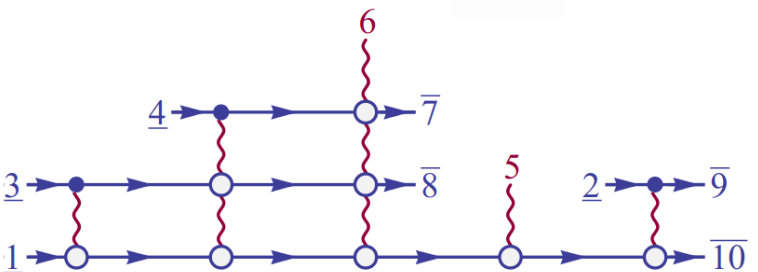
Colour-dressing Melia's amplitudes (**JO 1507.00332**)

$$\mathcal{A} = \sum \underline{C_i}\,\underline{A_i} \Longrightarrow (\underline{1}\,\underline{3}\,\underline{4}\,6\,\overline{7}\,\overline{8}\,5\,\underline{2}\,\overline{9}\,\overline{10}) \Leftrightarrow$$





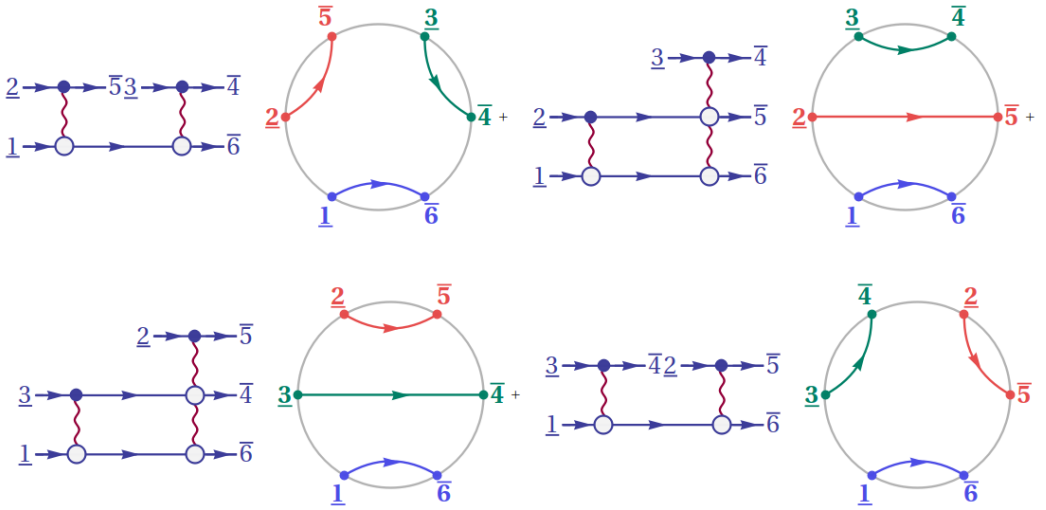
$\Longleftarrow$



$$= \{\underline{1}|\mathbf{T}_{a_1}\mathbf{T}_{a_2}\mathbf{T}^6\mathbf{T}^5\mathbf{T}_{a_3}|\overline{10}\}\{\underline{2}|\overline{\mathbf{T}}^{a_3}|\overline{9}\}\{\underline{3}|\overline{\mathbf{T}}^{a_1}|\overline{8}\}\{\underline{4}|\overline{\mathbf{T}}^{a_2}|\overline{7}\} + \cdots$$

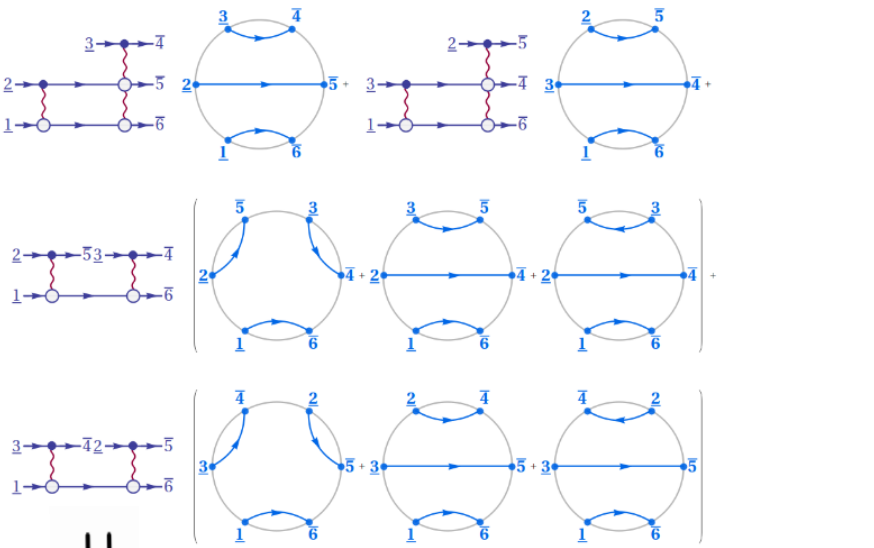
Colour-dressed QCD (**BPV 2604.18672**)

$$\mathcal{A}_3(1,2,\bar5,3,\bar4,\bar6)\,C[1,2,\bar5,3,\bar4,\bar6]^{ccc}{}_{ccc} + \mathcal{A}_3(1,2,3,\bar4,\bar5,\bar6)\,C[1,2,3,\bar4,\bar5,\bar6]^{ccc}{}_{ccc} +$$
$$\mathcal{A}_3(1,3,\bar4,2,\bar5,\bar6)\,C[1,3,\bar4,2,\bar5,\bar6]^{ccc}{}_{ccc} + \mathcal{A}_3(1,3,2,\bar5,\bar4,\bar6)\,C[1,3,2,\bar5,\bar4,\bar6]^{ccc}{}_{ccc}$$



$\Rightarrow$

$$\big(A_3(1,2,\bar5,3,\bar4,\bar6) + A_3(1,2,3,\bar5,\bar4,\bar6)\big)\,C[1,2,\bar5,3,\bar4,\bar6]^{ccc}{}_{ccc} + A_3(1,2,3,\bar4,\bar5,\bar6)\,C[1,2,3,\bar4,\bar5,\bar6]^{ccc}{}_{ccc} +$$
$$\big(A_3(1,3,\bar4,2,\bar5,\bar6) + A_3(1,3,2,\bar4,\bar5,\bar6)\big)\,C[1,3,\bar4,2,\bar5,\bar6]^{ccc}{}_{ccc} + A_3(1,3,2,\bar5,\bar4,\bar6)\,C[1,3,2,\bar5,\bar4,\bar6]^{ccc}{}_{ccc}$$



$\Downarrow$

SparseArray [Specified elements: 4548 Dimensions: {7, 7, 7, 7, 7, 7}], $\dfrac{[23]^2\langle56\rangle^2}{[34]\langle61\rangle\langle5|(16)|2]\,s_{561}} - \dfrac{[12]^2\langle45\rangle^2}{[61]\langle34\rangle\langle5|(16)|2]\,s_{612}}$

SparseArray [Specified elements: 3480 Dimensions: {7, 7, 7, 7, 7, 7}], $\dfrac{\langle5|(46)|3]^3}{[34]\langle25\rangle\langle1|(25)|3]\langle5|(12)|6]\,s_{125}} - \dfrac{[23]^2\langle46\rangle^2}{[35]\langle61\rangle\langle4|(16)|2]\,s_{461}} + \dfrac{[32]^3\langle46\rangle^2\langle1|(23)|5]}{[25][53]\langle61\rangle\langle1|(25)|3]\langle4|(16)|2]\,s_{461}} + \dfrac{[12]^2\langle54\rangle^2}{[61]\langle35\rangle\langle4|(16)|2]\,s_{612}} + \dfrac{[12]^2\langle54\rangle^3\langle3|(12)|6]}{[61]\langle34\rangle\langle53\rangle\langle4|(16)|2]\langle5|(12)|6]\,s_{612}}$

SparseArray [Specified elements: 4548 Dimensions: {7, 7, 7, 7, 7, 7}], $\dfrac{[32]^2\langle46\rangle^2}{[25]\langle61\rangle\langle4|(16)|3]\,s_{461}} - \dfrac{[13]^2\langle54\rangle^2}{[61]\langle25\rangle\langle4|(16)|3]\,s_{613}}$

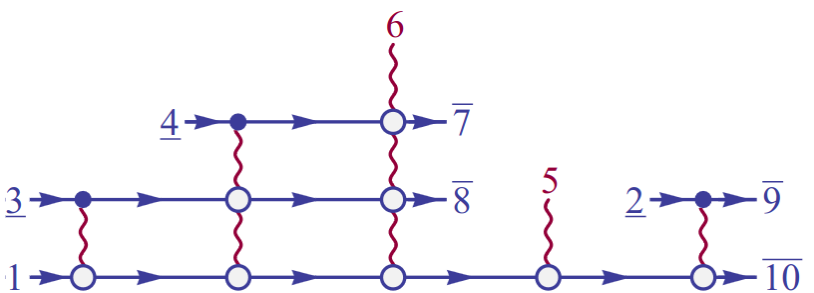
SparseArray [Specified elements: 3480 Dimensions: {7, 7, 7, 7, 7, 7}], $\dfrac{\langle4|(56)|2]^3}{[25]\langle34\rangle\langle1|(34)|2]\langle4|(13)|6]\,s_{134}} - \dfrac{[32]^2\langle56\rangle^2}{[24]\langle61\rangle\langle5|(16)|3]\,s_{561}} + \dfrac{[23]^3\langle56\rangle^2\langle1|(23)|4]}{[34][42]\langle61\rangle\langle1|(34)|2]\langle5|(16)|3]\,s_{561}} + \dfrac{[13]^2\langle45\rangle^2}{[61]\langle24\rangle\langle5|(16)|3]\,s_{613}} + \dfrac{[13]^2\langle45\rangle^3\langle2|(13)|6]}{[61]\langle25\rangle\langle42\rangle\langle4|(13)|6]\langle5|(16)|3]\,s_{613}}$

Colour-dressed QED (**BPV 2604.18672**)

From QCD to QED: simply take $\mathfrak{g} = \mathfrak{u}_1$!

$$(T^a_{\mathbf{R}})^i{}_j \to \{1\}$$
$$(\overline{T}^a_{\mathbf{R}})^i{}_j \to -\{1\}$$

The tensors become simple *integer weights* for QED

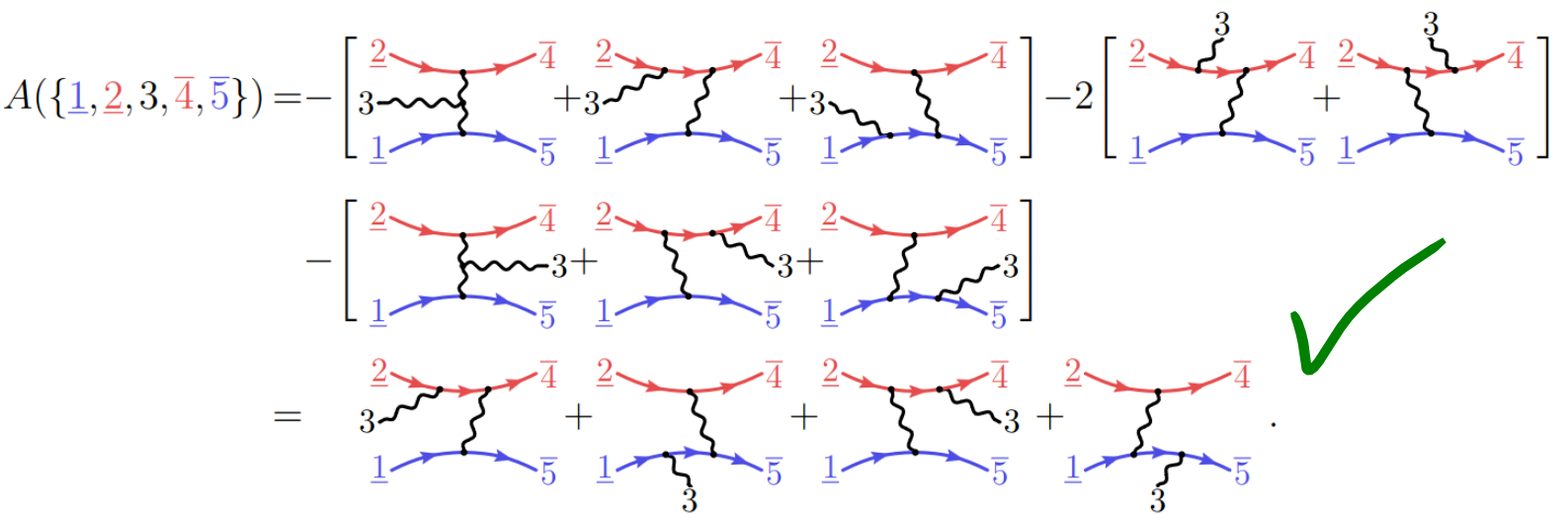


$\Longrightarrow \quad \mathcal{C}[\underline{1}\underline{3}\underline{4}6\overline{7}\overline{8}5\underline{2}\overline{9}\overline{10}] = 6$

Colour-dressed QED (**BPV 2604.18672**)

The job of these integers is to fix over-counting of Feynman diagrams that appear in multiple partial amplitudes.

$$A(\{\underline{1},\underline{2},3,\overline{4},\overline{5}\}) = -\mathcal{A}(\underline{1}3\underline{2}\overline{4}\overline{5}) - 2\mathcal{A}(\underline{1}\underline{2}3\overline{4}\overline{5}) - \mathcal{A}(\underline{1}\underline{2}\overline{4}3\overline{5})$$



Quod erat demonstrandum: QED!

Thanks!